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January 24, 1966
SUPERSEDING
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April 16, 1946

FEDERAL SPECIFICATION

HOSE, RUBBER (SLEEVES; DREDGING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dredging sleeves designed to withstand extreme abrasion and rough service in dredging operations.

1.2 Classification.

1.2.1 *Grade and type.* Dredging sleeves shall be the wrapped type, of a single grade.

1.2.2 *Sizes.* Unless otherwise specified (see 6.2), the sleeves furnished under this specification shall be of the following sizes: 6, 8, 10, 12, 16, 20, and 22 inches I.D. respectively.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specification:

CCC-C-419—Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-775—Hose, Rubber or Fabric (Including Tubing), and Fittings, Nozzles and Strainers, Packaging of.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Material not definitely specified

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shall be of the quality used for the purpose in commercial practice and shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2 Rubber. The rubber compound used in the tube and cover shall be manufactured from synthetic rubber, natural rubber, or a mixture of the two substances.

3.3 Cotton duck. The cloth shall be 30 ounces per square yard and conform to type I, CCC-C-419.

3.4 Preproduction sample. When specified (see 6.2), a preproduction sample shall be submitted for Government approval. The preproduction sample shall be the same grade, type, and size as the standard product submitted by the contractor to meet the requirements of this specification. Approval of preproduction sample authorizes commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities as used for the manufacture of production items. In addition, one linear yard, full width, of unfrictioned duck shall be submitted with the preproduction sample. The unfrictioned duck and the preproduction sample shall be inspected in accordance with 4.2.

3.5 Lengths. The length of the sleeve shall be as specified (see 6.2).

3.5.1 *Tolerances.* A tolerance of plus or minus one percent shall be permitted in the length of the sleeve.

3.6 Construction. The sleeves shall consist of a synthetic or natural rubber tube having layers of frictioned cotton duck. In addition, the sleeves shall have a layer of rubber between the two center reinforcements of the same compound as used in the tube, a synthetic or natural rubber cover, end caps, and a chafing strip.

3.6.1 *Duck.* Cotton duck shall be of square-woven design and in accordance with good manufacturing practice shall be free from unsightly defects, dirt, and knots.

3.6.1.1 *Plies.* The number of plies or layers of cotton duck shall be applied on a cut bias of forty-five degrees, with edges lapped at least one-half inch (not sewed), and shall be impregnated or frictioned with a rubber compound so that there is a distinct skim coat of rubber between the plies.

3.6.2 *End caps.* The ends of the sleeve shall be capped with the synthetic rubber compound used in the tube. The caps shall be not less than 1/4 inch in length.

3.6.3 *Chafing strips.* The sleeve shall have a chafing strip on each of its ends as protection against injury caused by connecting clamps. For a distance of 4, 6, or 8 inches from each end, as specified in the invitation for bids, the strip shall be a layer or ply of cotton duck conforming to 3.6.1.

3.7 Physical requirements. The hose shall meet the requirements of table I.

3.8 Identification marking. Embossed on each sleeve shall be the manufacturer's name or trademark, or both, and the month and year of the manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilitites or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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TABLE I. *Physical test requirements*

Sizes I.D. (in).	6	8	10	12	16	20	22
Plies (minimum)	6	6	8	10	12	14	14
Tolerances							
Diameter less than 12 inches	±1/8	±1/8	±1/8	—	—	—	—
Diameter over 12 inches	—	—	—	±1/4	±1/4	±1/4	±1/4
O.D. ends	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8
Thickness (inch)							
Tube	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Cover	1/16	1/16	3/32	3/32	3/32	1/8	1/8
Caps	1/4	1/4	1/4	1/4	1/4	1/4	1/4

Tensile strength
Before aging (min.)
Tube)
Cover) 2,000 lbs./sq. in.
After aging
Tube)
Cover) 1,500 lbs./sq. in.
Ultimate elongation
Tube before aging 500 percent after aging 375 percent
Cover before aging 500 percent after aging 375 percent
Tension set
After ten minutes rest, maximum 20 percent
Adhesion (lbs/inch of width) (minimum)
Tube and plies_____18 pounds
Cover and plies_____15 pounds
Between plies_____20 pounds

4.2 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall be tested at the place of manufacture. The preproduction sample shall be examined in accordance with 4.4 and tested in accordance with 4.5. The duck submitted (see 3.4) shall be examined in accordance with 4.4.1.

4.3 Sampling for lot acceptance.

4.3.1 *Inspection lot.* A lot shall consist of all sleeves of the same size offered for delivery at one time.

4.3.2 *Sampling for examination.* A random sample of dredging sleeves offered to the Government shall be selected from each lot in accordance with table I in section 6 of Fed. Test Method Std. No. 601.

4.3.3 *Sampling for test.* A full ring sample 1-1/2 inches wide shall be furnished for each sample from each lot in accordance with group B, table II, of section 6, Fed. Test Method Std. No. 601. Sample sleeves shall be selected at random during manufacture and built so that a full ring sample 1-1/2 inches wide can be cut from one end of the sleeve prior to fitting the end cap and end cover. After the end cap and end cover are attached to the sleeve, the ring sample and sleeve shall be vulcanized together, and tested in accordance with 4.5.

4.4 Examination and tests. Each of the sample lengths of dredging sleeves selected in accordance with 4.3.2 and 4.3.3 shall be examined to verify compliance with this specification.

4.4.1 *Duck.* The sample of duck furnished in accordance with 4.2 shall be examined for compliance with 3.6.1.

4.5 Tests.

4.5.1 *Test methods.*

4.5.1.2 *Physical tests.* The following tests shall be made on samples selected in ac-

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cordance with the applicable test methods specified in Fed. Test Method Std. No. 601.

4.5.1.3 *Diameter of sleeve.* The inside diameter of the sleeve shall be determind as described in method 2221 of Fed. Test Method Std. No. 601.

4.5.1.4 *Thickness.* The thickness of the tube, cover and caps shall be determined as described in method 2011 of Fed. Test Method Std. No. 601.

4.5.1.5 *Tensile strength and ultimate elongation.* The tensile strength and elongation of the tube and cover shall be determined as described in methods 4111 and 4121, respectively, of Fed. Test Method Std. No. 601. Die II shall be used for cutting the specimen.

4.5.1.6 *Accelerated aging.* The resistance of the tube and cover to accelerated aging shall be determined as described in method 7111 of Fed. Test Method Std. No. 601 except that the time of aging shall be 48 hours. Tensile strength and ultimate elongation tests shall be used to determine the amount of deterioration.

4.5.1.7 *Adhesion.* The adhesion between plies and between plies and rubber parts shall be determined as described in method 8011 of Fed. Test Method Std. No. 601.

4.5.1.8 *Tension set.* The tension set of the tube and cover shall be determined as described in method 4411 of Fed. Test Method Std. No. 601.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* The sleeves shall be preserved and packaged in accordance with the applicable level A requirements of MIL-H-775.

5.1.2 *Level C.* The sleeves shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. Supplier's standard practice may be used when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The sleeves shall be packed in accordance with the applicable level A requirements of MIL-H-775.

5.2.2 *Level B.* The sleeves shall be packed in accordance with the applicable level B requirements of MIL-H-775.

5.2.3 *Level C.* The sleeves shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Dredging sleeves are intended for use between sections of steel pipe on pontoon dredging lines.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size and length required (see 1.2.2 and 3.5).
(*c*) When preproduction sample is required (see 3.4).
(*d*) Length of chafing strips on each end (see 3.6.3).
(*e*) Place of testing preproduction sample, if different (see 4.2).
(*f*) Levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.3 Transportation descriptions. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Hose, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Motor:

Hose, rubber, reinforced with fabric. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO
Navy—YD
Air Force—69

Review activities:

Army—MO
Navy—YD
Air Force—69

User activities:

Army—CE
Navy—MC
Air Force—None

Preparing activity:

Navy—YD

CIVIL AGENCIES

Interest:

GSA-FSS
HEW
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the [illegible]ments. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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